

Ethereum Risk States as a Tail-Risk Switch for Art NFTs: Evidence from SuperRare

Chen Ziwen

Singapore University of Social Sciences

zwchen005@suss.edu.sg

Abstract

Art-NFT markets are thin and typically quoted and settled in Ethereum, so stress in the settlement asset may tighten liquidity and funding conditions in the downstream market. This paper asks whether observable Ethereum (ETH) risk states provide an ex-ante ranking of subsequent crash risk in a curated art-NFT marketplace. Using SuperRare sales aggregated to a daily price proxy (2021–2023), we sort days by (i) 7-day realized ETH volatility and (ii) the filtered high-volatility probability from a two-state Markov-switching model. Forward 30-day drawdown crashes are sharply monotone across state quartiles: a 30% USD crash rate rises from 9.9% to 38.8% from the lowest to highest volatility-probability quartile. Because crash windows overlap mechanically, conventional logit inference is overconfident; we therefore report main results as conservative linear probability models with Newey-West HAC errors and a moving-block bootstrap. Using a fully real-time state proxy, the block-bootstrap p-value for severe ETH-denominated crashes (40% drawdown) is 0.019. The signal is strongest during the 2022 market stress episode, consistent with ETH risk states activating downstream tail risks when settlement-asset stress is genuine. The settlement asset operates as a tail-risk switch for art NFTs, with limited corresponding mean-return predictability.

Keywords: Non-fungible tokens; Ethereum; tail risk; crash risk; settlement asset; realized volatility; regime switching.

JEL classification: C22; G12; G14.

1. Introduction

Art-NFT markets are thin and heterogeneous, and transactions are typically quoted and settled in Ethereum (ETH). Because each NFT is unique and markets lack centralised order books, price formation depends critically on the availability of marginal buyers who are themselves exposed to the settlement asset. Limits-to-arbitrage and liquidity-spiral mechanisms therefore imply state-dependent downside risk: when the settlement asset becomes volatile and funding conditions deteriorate, market depth can vanish and drawdowns can cluster (Shleifer and Vishny, 1997; Brunnermeier and Pedersen, 2009). Recent evidence confirms that NFT valuations are tightly linked to cryptocurrency conditions and speculative cycles (Dowling, 2022b; Anselmi and Petrella, 2023; Wang et al., 2022).

Existing studies examine NFT pricing determinants (Dowling, 2022a,b; Horky et al., 2022), crypto-NFT interrelatedness and spillovers (Karim et al., 2022; Xia et al., 2022), NFT portfolio diversification and mean-variance properties relative to crypto assets (Ko et al., 2022), and high-dimensional crypto early-warning systems that integrate on-chain metrics, social sentiment, and machine-learning models (Ke et al., 2025). By contrast, this paper studies whether the risk state of a single core asset matters precisely because that asset serves as the settlement medium of a thin downstream market. The contribution is therefore not generic return predictability, but an interpretable and real-time ranking of downstream left-tail crash risk implementable without on-chain data pipelines or high-dimensional models.

We document four empirical facts. First, state-sorted crash rates are sharply monotone: sorting days by the filtered high-volatility probability from a two-state Markov-switching model (Hamilton, 1989), the 30-day USD

crash rate rises from 9.9% (Q1) to 38.8% (Q4). Second, the same state sort predicts severe ETH-denominated crashes (40% drawdown: 7.6% to 27.6%), making a purely mechanical USD/ETH translation story less likely. Third, using a fully real-time state proxy free of parameter look-ahead, block-bootstrap p-values remain below 0.05 for severe ETH crash40 ($p = 0.019$), and a permutation test yields $p < 0.001$. Fourth, sub-period analysis shows the effect is strongest during the 2022 genuine market stress episode (ETH crash40 coefficient 0.350, $t = 2.56$), not the 2021 speculative boom, consistent with a settlement-asset transmission mechanism that activates when funding constraints bind.

The inference challenge is non-trivial. Overlapping forward crash definitions imply strong serial dependence in the crash indicator, so conventional heteroskedasticity-robust standard errors in logit regressions can be severely overconfident. We report main results using a linear probability model with Newey–West (1987) HAC standard errors and a moving-block bootstrap (Künsch, 1989), supplemented by a permutation test and non-overlapping subsample estimates.

The remainder of the paper is organised as follows. Section 2 describes the data and market proxy. Section 3 defines the ETH risk state measures. Section 4 presents the empirical strategy. Section 5 reports and discusses the results. Section 6 concludes.

2. Data and market proxy

Our art-NFT data come from SuperRare sales records, a curated digital-art marketplace where each work is unique (Nadini et al., 2021; Horky et al., 2022). Each record includes a transaction timestamp and a sale price in USD. We aggregate transactions to the daily level over 2021-04-16 to 2023-06-07, yielding 21,170 sales spanning 783 calendar days (mean 27.04 sales/day).

For each day t , we compute the daily median USD sale price as the primary price proxy P_t . The median is chosen because it is less sensitive than the mean to a few high-value sales in a thin market; a few extreme transactions cannot mechanically shift the median in the way they would shift the mean. We report robustness to alternative daily aggregators (mean, 10% trimmed mean, 5% winsorized mean), weekly aggregation, and minimum transaction-count subsamples. The daily mean/median ratio is essentially flat across ETH regime quartiles (range: 0.9998–1.0014), indicating no systematic compositional drift in high-volatility states.

To express outcomes in ETH units, we divide the daily USD proxy by the same-day ETH/USD close (CoinMarketCap), producing P_t^{ETH} . This denomination is central: SuperRare transactions in our sample are closely linked to ETH as the settlement asset, and crypto-art valuations are closely tied to that asset (Anselmi and Petrella, 2023).

3. Ethereum risk states

Our baseline state variable is the 7-day realized volatility vol7_t , computed as the rolling standard deviation of daily log ETH returns. A top-quartile indicator highvol_t equals one when vol7_t exceeds the full-sample 75th percentile.

As our primary state measure, we use a fully real-time indicator highvol_{rt} , equal to one when vol7_t exceeds the expanding-window 75th-percentile threshold of vol7 estimated using only data available through $t-1$. This measure is completely free of look-ahead bias in both level and threshold, and is our preferred measure for Table 1 Panel A.

As a secondary measure, we estimate a two-state Markov-switching model for ETH returns with state-dependent volatility (Hamilton, 1989) and extract the filtered probability of the high-volatility regime, $P_{\text{high_filt}}$. Although the filtered probability at date t uses only returns up to t , the model parameters are estimated on the full sample; we therefore treat this as a secondary, in-sample-calibrated measure (cf. Ardia et al., 2019).

We do not explicitly control for Ethereum gas fees or other blockchain frictions; to the extent these rise in stress periods, part of the predictive content of ETH risk states may operate through downstream trading conditions. We return to this point in the robustness discussion below.

4. Empirical strategy

Returns. For completeness, we also examine cumulative NFT returns using local projections (Jordà, 2005). Mean-return predictability is materially weaker and less stable than crash predictability, so the paper focuses on downside risk.

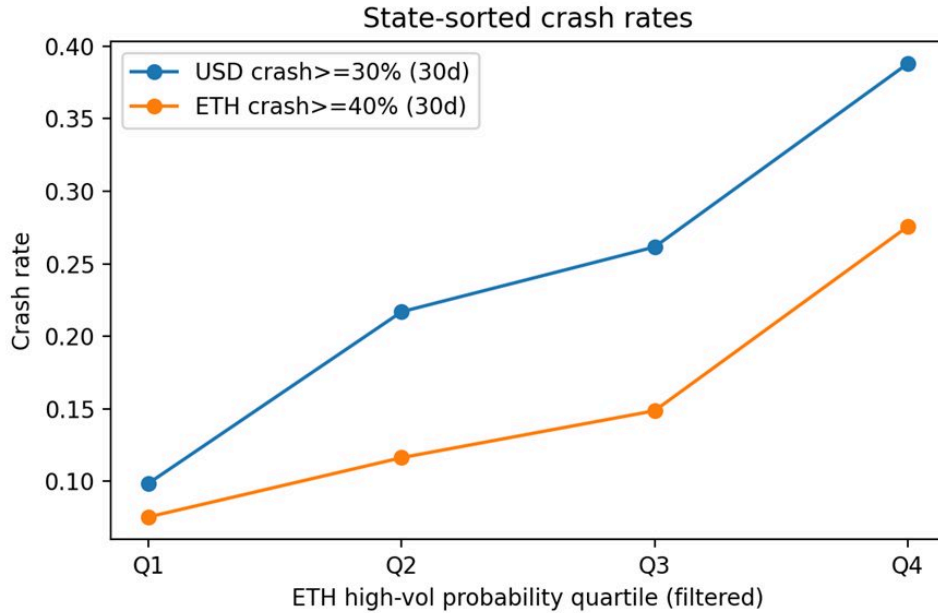
Crashes. For threshold $c \in \{0.2, 0.3, 0.4\}$ and horizon $H = 30$ days, we define $\text{Crash}_t(c) = 1\{\min_{j=1,\dots,H} P_{t+j} < (1-c)P_t\}$. The 30-day horizon reflects a standard monthly risk-management window for illiquid assets where portfolio adjustments are constrained by infrequent trading. The thresholds span moderate (20%), large (30%), and severe (40%) drawdowns, in line with the broader crypto crash-risk literature (Anastasiou et al., 2021), and allow us to assess whether ETH risk states primarily matter in the extreme left tail.

Because $\text{Crash}_t(c)$ and $\text{Crash}_{t+1}(c)$ share $H-1$ future days, the crash indicator is highly serially dependent. We emphasise conservative inference using: (i) a linear probability model with Newey-West (30) standard errors; (ii) a moving-block bootstrap with block length 30; (iii) a permutation test in which the ETH risk state is randomly shuffled 2,000 times to construct the finite-sample null distribution; and (iv) non-overlapping subsamples (every 30th day, $N \approx 25$) as a directional consistency check.

5. Results

5.1 State-sorted crash rates. Figure 1 shows crash rates sorted by quartiles of the filtered high-volatility probability $P_{\text{high_filt}}$. Crash risk rises monotonically with the ETH risk state. For a 30% USD drawdown, the 30-day crash rate increases from 9.9% (Q1) to 38.8% (Q4). For severe ETH-denominated crashes (40% drawdown), the rate increases from 7.6% (Q1) to 27.6% (Q4). This monotone pattern motivates our focus on tail risk.

Figure 1. State-sorted crash rates (30-day horizon).



Notes: The x-axis sorts days into quartiles of the filtered high-volatility regime probability $P_{\text{high_filt}}$. The blue line reports the 30-day crash rate for a 30% USD drawdown; the orange line reports the 30-day crash rate for a 40% ETH drawdown. $\text{Crash}_t(c) = 1\{\min_{j=1,\dots,30} P_{t+j} < (1-c)P_t\}$. Sample period: 2021-04-16 to 2023-05-08 (last 30 days excluded by construction for 30-day crash indicators; full sample 2021-04-16 to 2023-06-07).

5.2 Conservative crash inference and tail concentration. Table 1 reports strict inference results, organised in two panels. Panel A leads with the fully real-time state indicator `highvol_rt`, free of any parameter look-ahead. For USD-denominated 30% crashes, the real-time coefficient is 0.232 (block-bootstrap $p = 0.027$). For severe ETH-denominated 40% crashes, the coefficient is 0.203 (block-bootstrap $p = 0.019$). Panel B reports results using full-sample-calibrated indicators for comparison; results are directionally identical and statistically comparable. Across all specifications, block-bootstrap p -values for ETH crash40 are below 0.05. A permutation test in which the ETH risk state is randomly shuffled 2,000 times yields a one-sided p -value below 0.001 for ETH crash40, confirming the signal is not an artefact of the overlap structure.

Table 1. Crash probability and ETH risk states (strict inference)

Outcome (30-day crash)	State variable	LPM coef.	NW t-stat	Event rate	Boot CI 2.5%	Boot CI 97.5%	Boot p
Panel A. Primary results — real-time state indicator ($\dagger$)							
Crash30% (USD, 30-day)	<code>highvol_rt</code> $\dagger$	0.232	1.888	0.254	0.038	0.516	0.027
Crash40% (ETH, 30-day)	<code>highvol_rt</code> $\dagger$	0.203	2.011	0.143	0.041	0.441	0.019
Panel B. Comparison — full-sample state indicators							
Crash30% (USD, 30-day)	<code>P_high_filt</code>	0.294	1.721	0.246	0.009	0.703	0.045
Crash40% (ETH, 30-day)	<code>P_high_filt</code>	0.240	1.787	0.158	0.007	0.551	0.044
Crash30% (USD, 30-day)	<code>highvol</code>	0.196	2.008	0.248	0.025	0.412	0.024
Crash40% (ETH, 30-day)	<code>highvol</code>	0.184	2.379	0.150	0.033	0.347	0.016

Notes: $\dagger$ Panel A uses `highvol_rt`, a fully real-time indicator based on an expanding-window 75th-percentile threshold for `vol7` using data through $t-1$. Panel B reports full-sample-calibrated indicators for comparison. LPM includes contemporaneous ETH return (not shown). Newey-West standard errors use `maxlags = 30`. Moving-block bootstrap uses block length 30 and 2,000 replications; percentile CIs and two-sided p -values. Sample period: 2021-04-16 to 2023-05-08 (last 30 days excluded by construction for 30-day crash indicators; full sample 2021-04-16 to 2023-06-07).

Robustness to thin trading. With an average of 27 transactions per day, a potential concern is that day-to-day changes in the daily median may reflect shifts in the compositional mix of artworks rather than genuine price-level movements. Two diagnostics address this concern. First, the ratio of the daily mean to median price is flat across ETH regime quartiles (Q1: 1.0004; Q2: 1.0014; Q3: 0.9998; Q4: 1.0014), indicating no systematic compositional drift. Second, restricting to days with transaction counts above the top-half threshold ($tx \geq 25$) and top-tercile threshold ($tx \geq 30$) yields `highvol` coefficients of 0.213 ($t = 2.15$) and 0.232 ($t = 1.85$) for USD crash30, and 0.175 ($t = 2.22$) and 0.157 ($t = 1.72$) for ETH crash40 (Appendix Table A1). Taken together, these checks are inconsistent with thin-day composition being the primary driver of the crash results.

Robustness to overlap and low effective sample size. Because the forward crash indicator overlaps mechanically, we also report two complementary checks. The permutation test yields a one-sided p -value below 0.001 for ETH crash40 (Appendix Table A3). In a non-overlapping subsample (every 30th observation, $N = 25$; Appendix Table A2), all state coefficients remain positive across six specifications, confirming directional consistency despite the low power inherent at $N = 25$.

5.3 Sub-period analysis and state-contingent interpretation. Table 2 partitions the sample into the NFT boom (April–December 2021), the crash phase (January–June 2022), and the post-crash period (July 2022–June 2023). The ETH crash40 coefficient is smallest during the boom (0.147, $t = 1.43$) and strongest during the crash phase (0.350, $t = 2.56$), precisely the period of genuine ETH market stress. This pattern is theoretically coherent: limits-to-arbitrage and

liquidity-spiral mechanisms predict that settlement-asset risk regimes transmit to downstream tail risks when funding constraints genuinely bind, not during speculative upswings.

Post-crash predictability for ETH crash40 is attenuated (0.066, $t = 0.65$), reflecting the lower-volatility 2022–2023 environment. This is consistent with the signal functioning as a state-contingent risk overlay rather than an unconditional forecasting device: it ranks downstream crash risk conditional on the settlement asset entering a stress regime. When genuine ETH stress is absent, the signal appropriately carries limited information. USD crash30 remains positive and directionally consistent across all sub-periods (boom: +0.090; crash: +0.239; post-crash: +0.290).

Table 2. Sub-period analysis: crash probability by market phase

Sub-period	N	coef (USD30)	t-stat (NW30)	Event rate	coef (ETH40)	t-stat (NW30)	Event rate
<i>State variable: highvol (top-quartile 7-day realized volatility)</i>							
Full sample (Apr 2021–Jun 2023)	746	+0.196	2.01	0.248	+0.184	2.38	0.150
Boom (Apr 2021–Dec 2021)	253	+0.090	0.84	0.126	+0.147	1.43	0.103
Crash (Jan 2022–Jun 2022)	181	+0.239	1.51	0.586	+0.350*	2.56	0.320
Post-crash (Jul 2022–Jun 2023)	312	+0.290	1.61	0.151	+0.066	0.65	0.090

Notes: LPM estimates of Crash_{t(c)} on highvol and contemporaneous ETH return (not shown). Newey-West standard errors use maxlags = 30. * $p < 0.05$ (NW). Sub-period boundaries: Boom = NFT speculative expansion; Crash = broad crypto and ETH/NFT market stress; Post-crash = lower-volatility stabilisation. Sample period: 2021-04-16 to 2023-05-08 (last 30 days excluded by construction for 30-day crash indicators; full sample 2021-04-16 to 2023-06-07).

Macroeconomic robustness. A potential concern is that ETH risk states and NFT crashes are jointly shaped by broader macroeconomic conditions—notably, the 2022 tightening cycle, which raised ETH volatility and reduced risk appetite across asset classes. To address this, we add cumulative FOMC monetary policy surprises (MP1, from the U.S. Monetary Policy Event-Study Database; Acosta et al., 2025) and cumulative 2-year Treasury yield changes as controls. The highvol coefficient for ETH crash40 moves from 0.184 to 0.169 (NW t from 2.38 to 2.28), while the macro controls themselves are statistically insignificant (cumMP1: $t = 0.96$, $p = 0.34$; cumUST2Y: $t = -0.02$, $p = 0.98$). Full results are in Appendix Table A4. These estimates suggest that the ETH risk-state signal is not fully explained by the added macro controls, while still leaving open the possibility of broader common factors.

Tail concentration and mechanism. The tail concentration is consistent with threshold-based capitulation: as ETH enters a high-volatility regime, liquidity providers and marginal buyers may withdraw, and a small number of forced sales can push prices through drawdown thresholds. The minimum-transaction-count restrictions reported above are consistent with this channel: the effect survives after excluding the thinnest trading days, suggesting that the mechanism operates through valuation uncertainty and buyer retrenchment rather than purely through realised trading volume. The fact that ETH-denominated crashes are also predictable is consistent with this interpretation: because the settlement asset and the buyer base overlap, ETH stress compresses the funding capacity and risk tolerance of marginal buyers simultaneously, producing real valuation declines rather than a purely mechanical USD/ETH translation. Direct identification of the transmission channel using bid-ask spreads or order-book depth is beyond the scope of this paper and is left for future research.

6. Conclusion

Using SuperRare sales aggregated to a daily market proxy, this paper shows that ETH risk states provide an ex-ante, implementable ranking of crash risk in art NFTs. All primary results are established with a fully real-time state indicator free of parameter look-ahead. The effect is concentrated in the severe left tail, survives conservative inference addressing overlapping crash windows (permutation $p < 0.001$), holds in ETH-denominated prices, and is robust to compositional diagnostics and minimum transaction-count restrictions. Sub-period analysis reveals that crash predictability is strongest during the 2022 genuine market stress episode, consistent with a settlement-asset transmission mechanism that activates when funding constraints genuinely bind.

Two boundary conditions merit note. First, the daily median price proxy retains an inherent limitation in any thin, heterogeneous market, though the compositional diagnostics reported above are inconsistent with compositional shift as the primary driver. Second, post-crash predictability is attenuated, indicating the signal functions as a state-contingent risk overlay rather than an unconditional long-run predictor.

One limitation is that ETH risk states may partly reflect broader macro and crypto-wide stress conditions. While our results survive the addition of cumulative FOMC monetary policy surprise controls, disentangling ETH-specific settlement effects from market-wide stress remains an open identification challenge that we interpret cautiously. More broadly, the evidence comes from a single curated marketplace with a specific settlement structure; generalisability of the settlement-asset framework across platforms and settlement currencies remains to be established.

Future research could extend this framework in three directions. First, incorporating direct marketplace liquidity measures — such as high-frequency bid-ask spreads or order-book depth — would allow a more precise identification of the transmission channel. Second, applying the settlement-asset framework across multiple NFT platforms and settlement currencies (including non-ETH chains) would assess generalisability. Third, combining settlement-asset risk states with macro-liquidity proxies and monetary policy surprises in a unified model could help isolate the ETH-specific channel from broader funding-conditions effects.

For researchers, the results highlight a methodological pitfall: forward drawdown definitions require dependence-robust inference. For practitioners, the settlement asset can act as a tail-risk switch, suggesting that NFT risk limits and hedging should condition on ETH risk states, particularly during genuine market stress periods.

Data availability statement

The processed daily dataset and replication code will be made publicly available in an online repository (link to be added) prior to publication. ETH and BTC price data are from CoinMarketCap (accessed 2026-02-25). SuperRare transaction data were obtained via the SuperRare public API and third-party data aggregators.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT to assist with language editing and to improve clarity and organisation. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

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